

JEE MAIN REVISION MASTER: LOGARITHM

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	Logarithm — Standard Calculations & Problem Archetypes	4 PYQs	Quantitative Analysis
Standard	Logarithm — Core Principles & Governing Laws	2 PYQs	Core Principle
Standard	Logarithm — Applied Cases, Reactions & Traps	2 PYQs	Applied Topics

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Logarithm — Standard Calculations & Problem Archetypes	$\sin^{-1}(\frac{\alpha + \sqrt{\beta}}{2})$, $\log_{(x+1)}(2x^2+7x+5) + \log_{(2x+5)}(x+1)^2 - 4 = 0$, $\log_{(x+1)}(2x+5)(x+1) + 2\log_{(2x+5)}(x+1) = 4$
Logarithm — Core Principles & Governing Laws	$e^{5(\log_e x)^2 + 3} = x^8, x > 0$, $\log_{\frac{1}{2}} \sin x = 2 - \log_{\frac{1}{2}} \cos x $, $\log_{\frac{1}{2}} \sin x $
Logarithm — Applied Cases, Reactions & Traps	3) is _____, $\log_{(x+3)}(6x^2+28x+30) = 5 - 2\log_{(6x+10)}(x^2+6x+9)$, $\log_2 1 = \log_2 x$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Logarithm — Standard Calculations & Problem Archetypes (Moderate Yield — 4 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Logarithm.

■■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "Let $\alpha = (1/4) + (1/8) + (1/16) + \dots$ and $\beta = (1/3) + (1/9) + (1/27) + \dots$. Then the value ..." [Ans: C]
- [2023 Morning Shift] Q#1 (MCQ): "If the solution of the equation $\log_{(\cos x)} \cot x + 4 \log_{(\sin x)} \tan x = 1, x \in (0, (\pi/2))$, is $\sin^{-1}(\frac{\alpha + \sqrt{\beta}}{2})$, ..." [Ans: N/A]
- [2023 Morning Shift] Q#4 (MCQ): "The number of integral solutions x of $\log_{((x+(7/2))((x-7/2x-3)))^2} \geq 0$ is ..." [Ans: N/A]

3.2 Logarithm — Core Principles & Governing Laws (Standard Yield — 2 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Logarithm.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Logarithm.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Logarithm PYQs.

Representative Source PYQs:

- [2025 Morning Shift] Q#3 (MCQ): "The product of all solutions of the equation $e^5(\log_e x)^2 + 3 = x^8, x > 0$, is :..." [Ans: N/A]
- [2020 Shift] Q#3 (Numerical): "The number of distinct solutions of the equation $\log_{((1/2))} |\sin x| = 2 - \log_{((1/2))} |\cos x|$ in the interval $[0, 2\pi]$, is :..." [Ans: 8]

3.3 Logarithm — Applied Cases, Reactions & Traps (Standard Yield — 2 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Logarithm.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2026 Evening Shift] Q#2 (MCQ): "The sum of all the real solutions of the equation $\log_{((x+3))} (6x^2 + 28x + 30) = 5 - 2\log_{((6x+10))} (x^2 + 6x + 9)$ is equal to :..." [Ans: N/A]
- [2021 Morning Shift] Q#2 (Numerical): "The number of solutions of the equation $\log_4 (x - 1) = \log_2 (x - 3)$ is :..." [Ans: 1]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 4th April Evening Shift (Q#1)

Let $\alpha = (1/4) + (1/8) + (1/16) + \dots$ and $\beta = (1/3) + (1/9) + (1/27) + \dots$. Then the value of $(0.2)^{\log_{(\sqrt{5})}(\alpha)} + (0.04)^{\log_{(5)}(\beta)}$ is equal to :

- (A) 4
- (B) 5
- (C) 8 [Correct]
- (D) 25

Correct Answer: C

Solution Analysis: We first find the values of α and β . Given $\alpha = (1/4) + (1/8) + (1/16) + \dots$. This is an infinite geometric progression with $a = (1/4), r = (1/2)$. So, $\alpha = (a / (1 - r)) = ((1/4) / (1 - (1/2))) = ((1/4) / (1/2)) = (1/2)$. Now, $\beta = (1/3) + (1/9) + (1/27) + \dots$. This is also an infinite geometric progression with $a = (1/3), r = (1/3)$. Thus, $\beta = (a / (1 - r)) = ((1/3) / (1 - (1/3))) = ((1/3) / (2/3)) = (1/2)$. So both are equal to $(1/2)$. Now the expression becomes

Benchmark Problem 2: JEE Main 2026 (Online) 23rd January Evening Shift (Q#2)

The sum of all the real solutions of the equation $\log_{((x+3))}(6x^2+28x+30)=5-2\log_{((6x+10))}(x^2+6x+9)$ is equal to :

- (A) 1
- (B) 4
- (C) 0
- (D) 2

Correct Answer: None

Solution Analysis: $\log_{((x+3))}(6x^2+28x+30)=5-2\log_{((6x+10))}(x^2+6x+9)-1$

$$\log_{((x+3))}2(3x^2+14x+15)+2\log_{((6x+10))}(x^2+6x+9)=5 \log_{((x+3))}2(x+3)(3x+5)+2\log_{((6x+10))}(x+3)^2=5$$

$$\log_{((x+3))}(x+3)+\log_{((x+3))}(6x+10)+4\log_{((6x+10))}(x+3)=5 \log_{((x+3))}(6x+10)+4\log_{((6x+10))}(x+3)=4\log_{((x+3))}(x+3)=1$$

$$\text{Let, } \log_{((x+3))}(6x+10)=t \Rightarrow \log_{((6x+10))}(x+3)=(1/t) \therefore t+4 \times (1/t)=4 \quad t^2-4t+4=0 \quad (t-2)^2=0 \quad t=2$$

$$\text{Checking } x=1 \text{ in equation (1)} \quad \& \log_{((1+3))}(6(1))=6(1)$$

Benchmark Problem 3: JEE Main 2025 (Online) 22nd January Morning Shift (Q#3)

The product of all solutions of the equation $e^5(\log_e x)^2+3=x^8, x>0$, is :

- (A) e^2
- (B) e
- (C) $e^{6/5}$
- (D) $e^{8/5}$

Correct Answer: None

Solution Analysis: We begin with the equation: $e^5(\log_e x)^2+3=x^8, x>0$. Equating the exponents, we have:

$5(\log_e x)^2+3=\log_e x^8=8\log_e x$. Let $t=\log_e x$. Substituting, the equation becomes: $5t^2+3=8t$. Rewriting this as a quadratic equation: $5t^2-8t+3=0$. Factoring the quadratic: $5t^2-5t-3t+3=0$, $(5t-3)(t-1)=0$. Thus, the solutions for t are: $t=1$ implies $\log_e x=1$, giving $x=e$. $t=(3/5)$ implies $\log_e x=(3/5)$, giving $x=e^{(3/5)}$. The product of all solutions is: $e^1 \times e^{(3/5)}=e^{1+(3/5)}=e^{(8/5)}$.